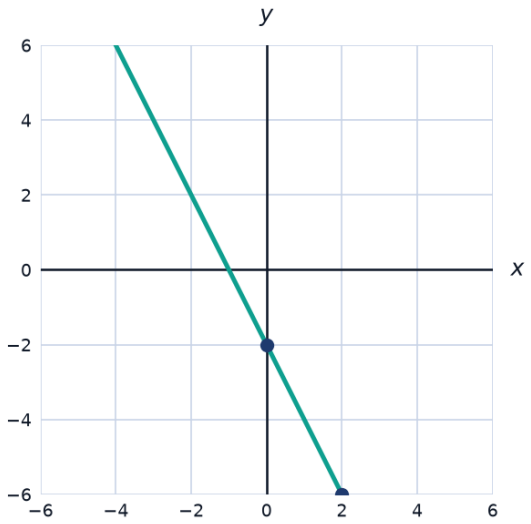
Desmos Graphing Tactics, Regression & Systems Strategy

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function f is shown. What is the value of $f(2)$?



2 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

What is the positive x-intercept of the graph of $y = x^2 + 3x - 4$ in the xy-plane?

3 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

What is the positive x-intercept of the graph of $y = x^2 + x - 6$ in the xy-plane?

4 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of $y = x$ and $y = -x - 2$ intersect at the point (a, b) in the xy-plane. What is the value of a?

5 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of $y = 4x - 2$ and $y = -2x + 16$ intersect at the point (a, b) in the xy-plane. What is the value of a?

6 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graphs of $y = x + 3$ and $y = -3x + 7$ intersect at the point (a, b) in the xy-plane. What is the value of a?

7 [Foundational] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

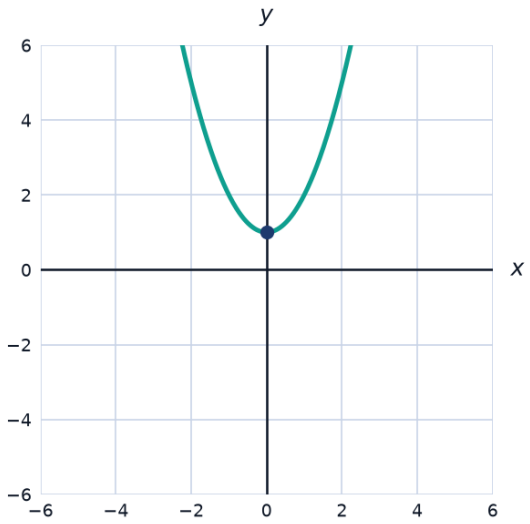
The graphs of $y = 4x$ and $y = -3x - 21$ intersect at the point (a, b) in the xy-plane. What is the value of a?

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Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

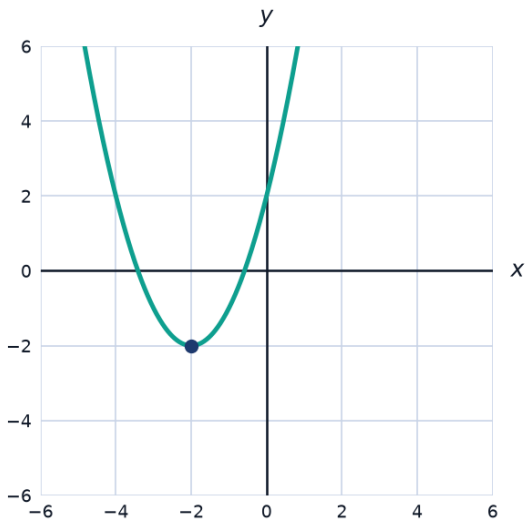
8 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x - h)^2 + k$, where h and k are constants, is shown. What is the value of $h + k$?



9 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

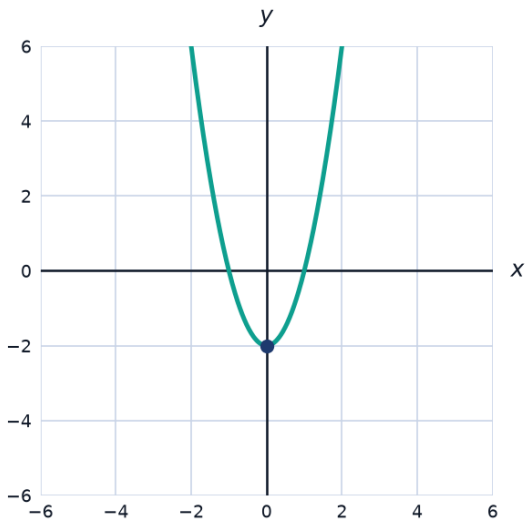
The graph of $y = (x - h)^2 + k$, where h and k are constants, is shown. What is the value of $h + k$?



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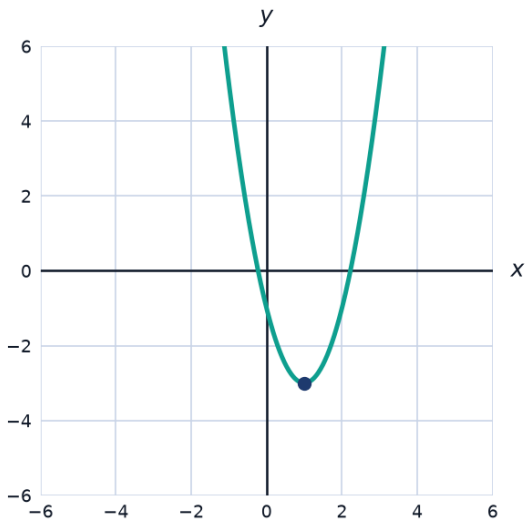
10 **[Intermediate]** • **[SPR]** • **Target: Graphing Intersections, Slider Constants & Desmos Modeling**

The graph of $y = 2(x - h)^2 + k$, where h and k are constants, is shown. What is the value of $h + k$?



11 **[Intermediate]** • **[SPR]** • **Target: Graphing Intersections, Slider Constants & Desmos Modeling**

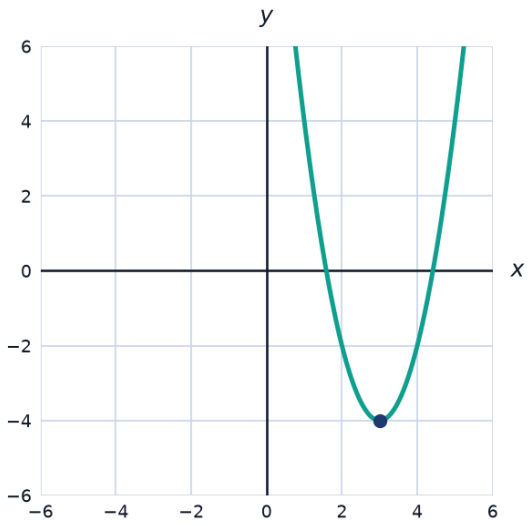
The graph of $y = 2(x - h)^2 + k$, where h and k are constants, is shown. What is the value of $h + k$?



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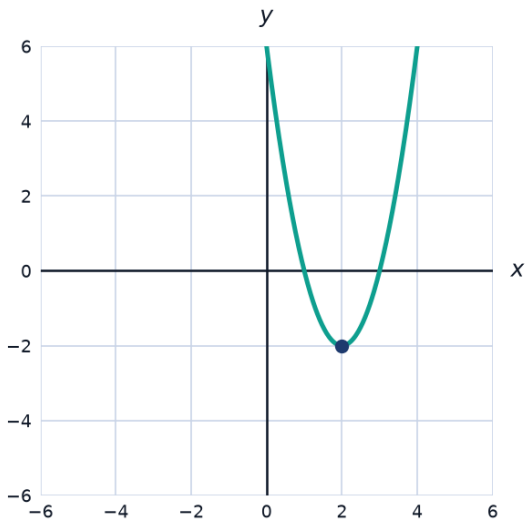
12 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = 2(x - h)^2 + k$, where h and k are constants, is shown. What is the value of $h + k$?



13 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = 2(x - h)^2 + k$, where h and k are constants, is shown. What is the value of $h + k$?



14 [Intermediate] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation of a circle in the xy-plane is $(x + 5)^2 + (y - 2)^2 = 9$. What is the radius of the circle?

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, the line $y = 4x - c$ and the parabola $y = x^2 - 3$ intersect at exactly one point, where c is a constant. What is the value of c?

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16 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 - 8x + 6y - 39 = 0$ defines a circle in the xy-plane. What is the radius of the circle?

17 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x + 4)(x - 3)$ intersects the x-axis at the points $(p, 0)$ and $(q, 0)$. What is the value of $q - p$, where $q > p$?

18 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 + 10x - 2y - 38 = 0$ defines a circle in the xy-plane. What is the radius of the circle?

19 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 + 4x + 2y - 59 = 0$ defines a circle in the xy-plane. What is the radius of the circle?

20 [Challenging] • [SPR] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

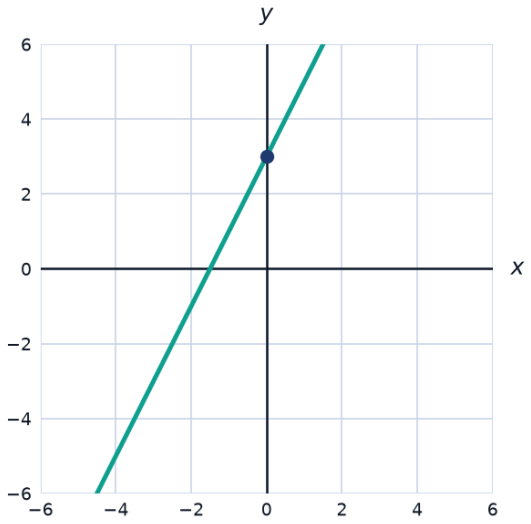
The equation $x^2 + y^2 - 8x + 10y - 23 = 0$ defines a circle in the xy-plane. What is the radius of the circle?

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function f is shown. What is the value of $f(2)$?



a. 3

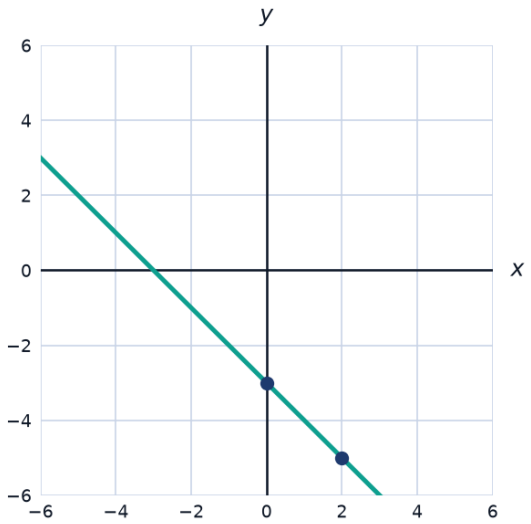
b. 2

c. -7

d. 7

22 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function f is shown. What is the value of $f(2)$?



a. -1

b. 5

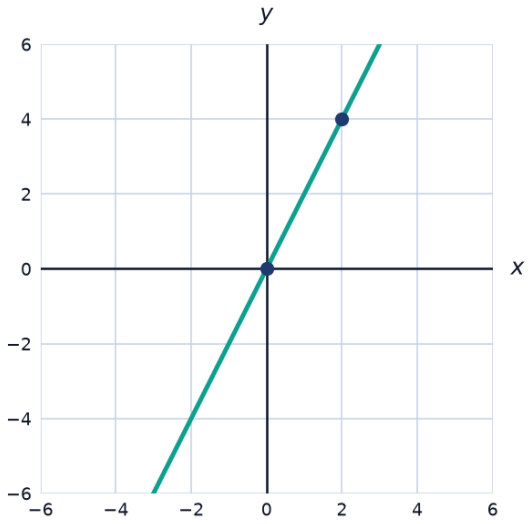
c. -5

d. -3

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23 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function f is shown. What is the value of $f(2)$?



a. 2

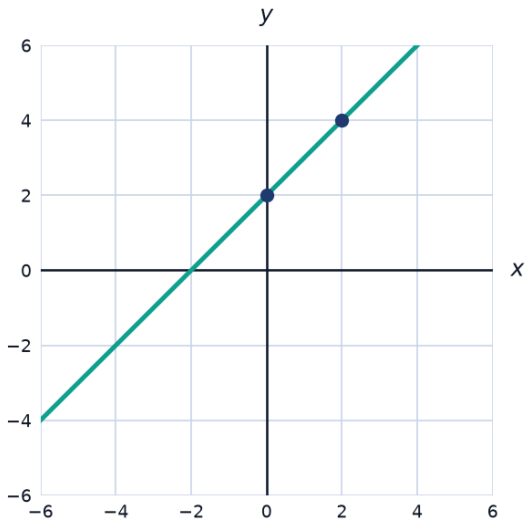
b. -4

c. 4

d. 0

24 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function f is shown. What is the value of $f(2)$?



a. 2

b. 1

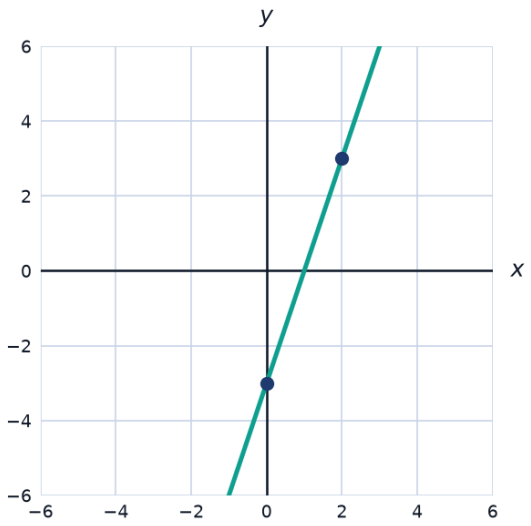
c. -4

d. 4

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25 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function f is shown. What is the slope of the graph of $y = f(x)$?



a. 5

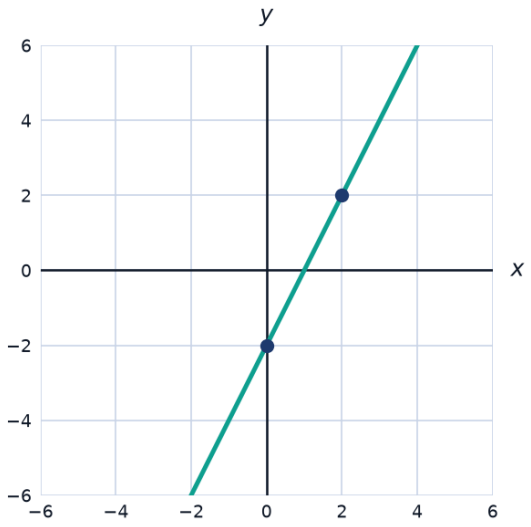
b. 3

c. −3

d. 4

26 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function f is shown. What is the slope of the graph of $y = f(x)$?



a. 2

b. −2

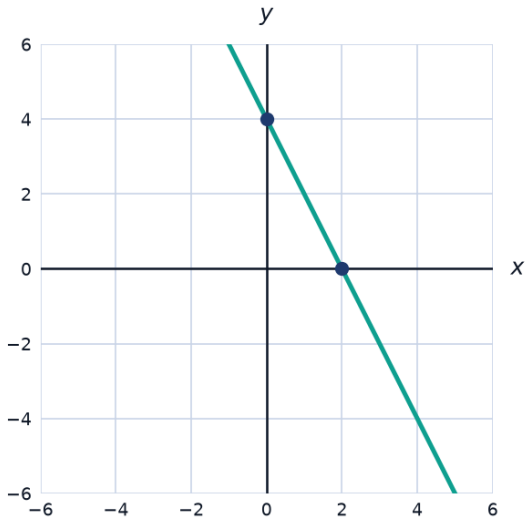
c. 3

d. 4

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27 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function f is shown. What is the slope of the graph of $y = f(x)$?



a. 2

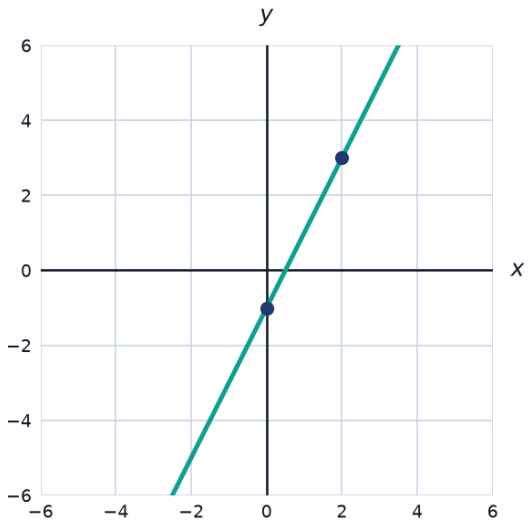
b. −2

c. 4

d. 0

28 [Foundational] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of the linear function f is shown. What is the slope of the graph of $y = f(x)$?



a. 3

b. −2

c. 2

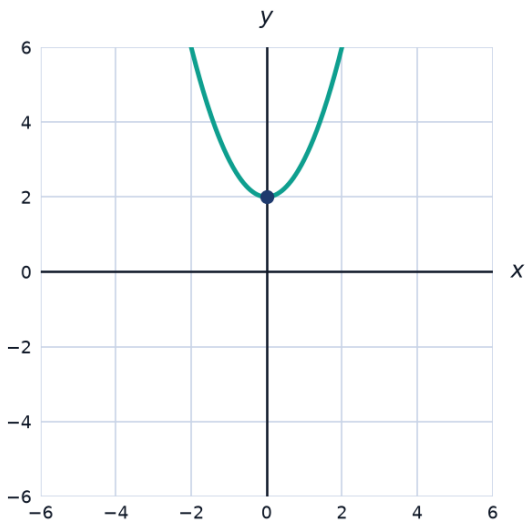
d. −1

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Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 **[Intermediate]** • **[MCQ]** • **Target: Graphing Intersections, Slider Constants & Desmos Modeling**

The graph of $y = (x - h)^2 + k$, where h and k are constants, is shown. What is the value of $h + k$?



a. 4

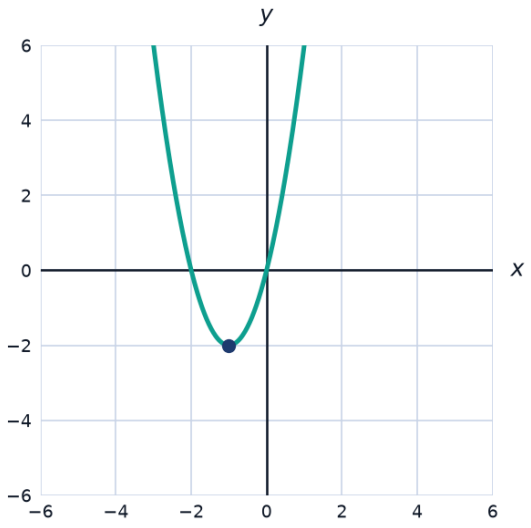
b. 2

c. −2

d. 0

30 **[Intermediate]** • **[MCQ]** • **Target: Graphing Intersections, Slider Constants & Desmos Modeling**

The graph of $y = 2(x - h)^2 + k$, where h and k are constants, is shown. What is the value of $h + k$?



a. 1

b. −1

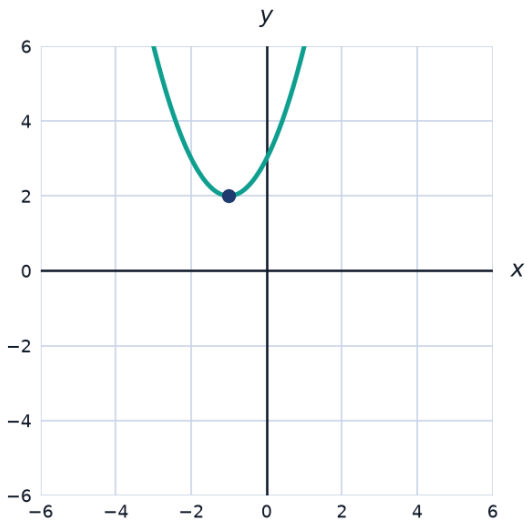
c. 2

d. −3

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31 **[Intermediate]** • **[MCQ]** • **Target: Graphing Intersections, Slider Constants & Desmos Modeling**

The graph of $y = (x - h)^2 + k$, where h and k are constants, is shown. What is the value of $h + k$?



a. -2

b. 1

c. -3

d. 3

32 **[Intermediate]** • **[MCQ]** • **Target: Graphing Intersections, Slider Constants & Desmos Modeling**

How many points of intersection do the graphs of $y = x^2 + 9$ and $y = 4x$ have in the xy-plane?

a. 3

b. 0

c. 1

d. 2

33 **[Intermediate]** • **[MCQ]** • **Target: Graphing Intersections, Slider Constants & Desmos Modeling**

How many points of intersection do the graphs of $y = x^2 + 2$ and $y = 2x$ have in the xy-plane?

a. 0

b. 1

c. 2

d. 3

34 **[Intermediate]** • **[MCQ]** • **Target: Graphing Intersections, Slider Constants & Desmos Modeling**

The equation of a circle in the xy-plane is $(x + 1)^2 + (y + 2)^2 = 16$. What is the radius of the circle?

a. 4

b. 16

c. 8

d. 5

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35 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

How many points of intersection do the graphs of $y = x^2 + 1$ and $y = 3x$ have in the xy-plane?

a. 1

b. 3

c. 2

d. 0

36 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation of a circle in the xy-plane is $(x - 4)^2 + (y - 2)^2 = 16$. What is the radius of the circle?

a. 4

b. 16

c. 8

d. 5

37 [Intermediate] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

How many points of intersection do the graphs of $y = x^2 + 7$ and $y = 2x$ have in the xy-plane?

a. 0

b. 1

c. 2

d. 3

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, the line $y = 2x - c$ and the parabola $y = x^2 + 4$ intersect at exactly one point, where c is a constant. What is the value of c?

a. 4

b. -4

c. -3

d. 3

39 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x + 4)(x - 2)$ intersects the x-axis at the points $(p, 0)$ and $(q, 0)$. What is the value of $q - p$, where $q > p$?

a. 2

b. 6

c. -2

d. -8

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40 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 + 4x - 6y - 23 = 0$ defines a circle in the xy-plane. What is the radius of the circle?

a. 8

b. 6

c. 36

d. 23

41 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = (x + 2)(x - 1)$ intersects the x-axis at the points $(p, 0)$ and $(q, 0)$. What is the value of $q - p$, where $q > p$?

a. 3

b. -1

c. -2

d. 1

42 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 - 6x + 8y - 39 = 0$ defines a circle in the xy-plane. What is the radius of the circle?

a. 64

b. 39

c. 10

d. 8

43 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

In the xy-plane, the line $y = 4x - c$ and the parabola $y = x^2 - 4$ intersect at exactly one point, where c is a constant. What is the value of c?

a. -8

b. 16

c. 4

d. 8

44 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The graph of $y = -(x)(x - 1)$ intersects the x-axis at the points $(p, 0)$ and $(q, 0)$. What is the value of $q - p$, where $q > p$?

a. 0

b. 3

c. -3

d. 1

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45 [Challenging] • [MCQ] • Target: Graphing Intersections, Slider Constants & Desmos Modeling

The equation $x^2 + y^2 - 2x - 3 = 0$ defines a circle in the xy-plane. What is the radius of the circle?

a. 4

b. 3

c. -2

d. 2

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